From Washington to Beijing: Time-Varying Transmission of Foreign Shocks to Brazil in a Trade-Weighted Block-Exogenous SVAR

[Author names]

Department of Economics, California State University, Fullerton

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Abstract

China overtook the United States as Brazil's largest trading partner in 2009. A constant-coefficient vector autoregression estimated across that transition averages two different economies. We estimate a block-exogenous structural VAR for Brazil in which the domestic block's loading on U.S. and Chinese activity is a linear function of each partner's lagged, smoothed share of Brazilian goods trade. On monthly data from 1999:07 to 2019:12, with four lags, a one-standard-deviation shock to Chinese activity raises Brazilian industrial production by 1.80% at its peak when trade weights are held at their 2019 values, against 0.69% at 2000 values. The 2019-minus-2000 difference, bootstrapped draw by draw, excludes zero at horizons of 6 to 24 months. A bootstrap likelihood-ratio test rejects the hypothesis that trade weights leave transmission unchanged ($p = 0.005$), and Chow and sup-LR tests reject parameter constancy in the constant model. The U.S. effect did not shrink with the U.S. trade share, consistent with financial rather than trade transmission. Block exogeneity of the foreign block is not rejected by a bootstrap lag-augmented Wald test at four and six lags. A kernel time-varying SVAR supports the China result. A Primiceri (2005) TVP-VAR with stochastic volatility attributes most of the change to volatility. All estimators are checked against simulated known answers, reference software and published results before use, and every number is reproducible from the open-source package.

Keywords: structural VAR; block exogeneity; time-varying parameters; trade linkages; China; Brazil; small open economy.
JEL: C32, E32, F41, F44.

1 Introduction

Small open economies import business cycles. How strongly they do so depends on who they trade with, and for Brazil that has changed dramatically. In 2000 the United States accounted for about 22% of Brazil's goods trade and China for less than 2%. By 2019 the order had reversed: China took 21%, the United States 14%. Most of Brazil's soybean and iron-ore exports now go to China.

Structural VARs (SVARs), introduced by Sims (1980), are the standard tool for measuring how foreign shocks transmit to a small open economy (Cushman and Zha, 1997; Canova, 2005; Maćkowiak, 2007). The usual specification assumes the transmission coefficients are constant over the sample. When the composition of trade shifts this much, a constant model estimates an average of the pre- and post-shift economies and may miss the shift entirely. Cesa-Bianchi, Pesaran, Rebucci and Xu (2012) addressed the question with a global VAR (GVAR) whose trade weights change over time. They found that China's

growing importance raised the long-run effect of Chinese GDP shocks on Latin America, while the U.S. effect fell less than its trade share.

This paper asks the same question in a single-country, block-exogenous SVAR that is light enough for a short monthly sample, and adds formal tests of whether the time variation is real. Our contributions are four:

1. **A trade-weighted block-exogenous SVAR.** The domestic block's loadings on each partner's activity are linear in that partner's lagged trade weight, following the interacted-panel approach of Towbin and Weber (2013). It nests a constant model and a GVAR-style "star" model. The foreign block is exogenous at every date and in every bootstrap draw.
2. **Tests of time variation.** We test whether time variation is present (Chow, sup-LR) and whether trade weights explain it (bootstrap LR), rather than imposing it.
3. **A ladder of robustness models.** These range from a nonparametric kernel SVAR (Giraitis, Kapetanios and Yates, 2014) to a Bayesian TVP-VAR with stochastic volatility (Primiceri, 2005; Del Negro and Primiceri, 2015).
4. **A replication pipeline.** Each estimator must recover known answers in simulation, match reference software, and reproduce published findings before it is applied to Brazil. Code, data hashes and every output are public (Section 9).

The main finding is that Brazil's industrial production now responds roughly two and a half times as strongly at its peak to Chinese activity shocks as it would have with 2000 trade linkages. The U.S. effect has not declined in proportion to the U.S. trade share.

2 Theoretical framework

2.1 A small open economy facing several partners

Consider a small open economy (SOE) that trades with partners $j = 1, \dots, J$. Following the small-open-economy tradition (Mendoza, 1995; Galí and Monacelli, 2005), the SOE takes foreign output, foreign interest rates and world commodity prices as given: its own shocks do not move them. This *block exogeneity* is the identifying restriction that separates foreign from domestic variables (Cushman and Zha, 1997; Zha, 1999).

Let home exports be the sum of exports to each partner, and let each partner's import demand depend on its activity $y^{*}_{j,t}$ with elasticity η_j. Log-linearising around a steady state in which partner j takes a share w_j of home trade gives

$$\widehat{EX}_t \;=\; \sum_{j=1}^{J} w_j \, \eta_j \, \hat{y}^{*}_{j,t} \;+\; \eta_q \, \hat{q}_t, \tag{1}$$

where hats denote log deviations and q_t is the real exchange rate. Through the trade channel, the elasticity of home demand with respect to partner j's activity is *proportional to the trade share* w_j. If w_j drifts, so does the transmission of partner-j shocks. Frankel and Rose (1998) document the empirical counterpart: trade intensity raises business-cycle co-movement.

2.2 Channels that do not scale with trade

Not all transmission runs through bilateral exports. Two other channels matter for Brazil:

- **Commodity prices.** Chinese demand raises world prices of soybeans, iron ore and oil, which improves Brazil's terms of trade whoever the buyer is (Fernández, Schmitt-Grohé and Uribe, 2017). We include a world commodity price index as a separate foreign variable so that this channel is not attributed to Chinese activity.
- **Financial conditions.** U.S. activity and U.S. monetary policy move global risk appetite, the dollar and capital flows to emerging markets (Canova, 2005; Maćkowiak, 2007). These effects scale with Brazil's financial exposure, not its share of goods trade with the United States.

We therefore model the loading of the domestic block on partner j's activity as an affine function of its trade weight:

$$\mathcal{B}_j(w_{j,t}) \;=\; B_j + w_{j,t}\, G_j. \tag{2}$$

G_j captures the trade channel in (1). B_j captures channels that do not scale with bilateral trade. Equation (2) nests a constant SVAR ($G_j = 0$) and a GVAR "star" loading ($B_j = 0$, $G_j = G$ for all j), in which the domestic block reacts to a single trade-weighted foreign aggregate $x^*_t = \sum_j w_{j,t} x_{j,t}$ (Pesaran, Schuermann and Weiner, 2004; Dees, di Mauro, Pesaran and Smith, 2007).

2.3 Testable hypotheses

H1 (rising China effect). G_{cn} is positive for Brazilian output, so the response to a Chinese activity shock is larger at 2019 than at 2000 trade weights.

H2 (trade-only U.S. channel). If U.S. shocks reach Brazil mainly through trade, the U.S. response falls as w_{us} falls. If financial channels dominate, it does not.

H3 (time variation). A constant-coefficient SVAR is rejected against one whose transmission moves with trade weights.

H4 (block exogeneity). Brazilian variables do not Granger-cause the foreign block.

3 Data

3.1 Macroeconomic series

All series are monthly. The foreign block contains world commodity prices, U.S. activity, Chinese activity and the federal funds rate. The domestic block contains Brazilian industrial production, consumer prices, the Selic policy rate and the real–dollar exchange rate (Table 1).

U.S. and Chinese activity use the same OECD measure, the normalised GDP reference series of the composite leading indicator system, so the two partners are on a common scale and can be combined with trade weights. No public source carries a monthly or quarterly real GDP index for China before 2011. A "shock to Chinese activity" is therefore a one-standard-deviation innovation to this detrended cycle, not to GDP growth.

Table 1. Variables, sources and transformations. Logs are multiplied by 100. Specification file: `examples/brazil/data.toml`.

Block	Variable	Description	Source (series)	Transform
Foreign	`commodity`	IMF all-commodity price index	FRED PALLFNFINDEXM	100×log
Foreign	`us_act`	U.S. activity, OECD GDP reference series, normalised	FRED USALORSGPNOSTSAM	level
Foreign	`cn_act`	China activity, same OECD measure	FRED CHNLORSGPNOSTSAM	level
Foreign	`ffr`	Effective federal funds rate	FRED FEDFUNDS	level
Domestic	`br_ip`	Industrial production, SA	IMF Production Indexes (PI)	100×log
Domestic	`br_cpi`	Consumer price index, all items, NSA	FRED BRACPIALLMINMEI	100×log
Domestic	`br_selic`	Overnight interbank (Selic) rate	FRED IRSTCI01BRM156N	level
Domestic	`br_er`	Reais per U.S. dollar, period average	IMF Exchange Rates (ER)	100×log

3.2 Trade weights

Bilateral trade comes from the IMF International Merchandise Trade Statistics (IMTS): goods exports (FOB) plus imports (CIF) by partner, with world trade as the total. Partner j's raw share in year y is $s_{j,y} = (X_{j,y} + M_{j,y})/(X_{W,y} + M_{W,y})$. Following Cesa-Bianchi et al. (2012), we smooth with a three-year average and lag it one year, so the weight used in every month of year y is fixed before that year's shocks:

$$w_{j,t} \;=\; \tfrac{1}{3}\sum_{k=1}^{3} s_{j,\,y(t)-k}. \tag{3}$$

Table 2. Trade weights at the three benchmark years (three-year average of the share of Brazil's goods trade, lagged one year).

Year	United States	China
2000	21.8%	1.8%
2009	15.8%	8.5%
2019	14.3%	20.6%

3.3 Sample

The headline sample runs from 1999:07 to 2019:12. Brazil floated the real in January 1999 and adopted inflation targeting in June 1999, so earlier data belong to a different monetary regime. On the longer 1996–2023 sample, every residual series rejects normality because of 2020, and pandemic observations dominate the estimates. We report the sample through 2023:11 as a robustness check (Section 6.4). With four lags, M1 is estimated on 242 monthly observations (1999:11–2019:12).

4 Econometric methodology

4.1 Block-exogenous SVAR (M0)

Stack the foreign variables x_t (dimension $n_x = 4$) and domestic variables z_t ($n_z = 4$) as $y_t = (x_t', z_t')'$. The block-exogenous VAR of order p is

$$
\begin{aligned}
x_t &= c_x + \sum_{l=1}^{p} A_{xx,l}\, x_{t-l} + u_{x,t}, \\
z_t &= c_z + \sum_{l=1}^{p} A_{zx,l}\, x_{t-l} + \sum_{l=1}^{p} A_{zz,l}\, z_{t-l} + u_{z,t},
\end{aligned}
\tag{4}
$$

so domestic lags are excluded from the foreign equations ($A_{xz,l} = 0$ for all l). Structural shocks satisfy $u_t = P\varepsilon_t$ with P lower triangular, foreign variables ordered first (commodity prices, U.S. activity, Chinese activity, federal funds rate), then Brazilian output, prices, policy rate and exchange rate. The zero upper-right block of P means foreign variables never respond to domestic shocks, on impact or later. Deterministic terms are a constant and, where noted, a linear trend. Each block is estimated by OLS.

Stability is tested with a Chow (1960) likelihood-ratio test at 2009:01, the month China overtook the United States, and with Andrews' (1993) sup-LR statistic over all admissible break dates. Both tests apply to the domestic block's coefficients, and p-values are bootstrapped.

4.2 Trade-weighted SVAR (M1)

The foreign block is as in (4). The domestic block lets partner variables enter through the trade-weighted loadings of (2):

$$
z_t = c + \sum_{l=0}^{p} \Big[D_l\, g_{t-l} + \sum_{j \in \{us,cn\}} \big(B_{j,l} + w_{j,t}\, G_{j,l} \big)\, \tilde{x}_{j,t-l} \Big] + \sum_{j} \phi_j\, w_{j,t} + \sum_{l=1}^{p} C_l\, z_{t-l} + e_t, \tag{5}
$$

where g_t collects the global variables (commodity prices, federal funds rate), $x_{j,t}$ is partner j's activity, and $\tilde{x}_{j,t} = x_{j,t} - \bar{x}_j$ is centred so that the product $w_{j,t}\tilde{x}_{j,t}$ cannot act as a disguised trend. The weight levels enter on their own through ϕ_j. The $l = 0$ terms are the contemporaneous foreign effects that block exogeneity allows. We call (5) the *interacted* variant. The *star* variant restricts $B_{j,l} = 0$ and $G_{j,l} = G_l$. The *levels* variant sets $G_{j,l} = 0$ and is the null model for H3. Estimation is equation-by-equation OLS with $w_{j,t}$ treated as predetermined.

Frozen-weight impulse responses. Given a fixed weight vector, (5) is a linear constant-coefficient VAR. For each benchmark year $s \in \{2000, 2009, 2019\}$ we set $w_{j,t} = w_{j,s}$, build the companion form, and compute impulse responses $\Phi_h(s)$ at horizon h. This is the counterfactual of Cesa-Bianchi et al. (2012): the same estimated structure, with the trade linkages of year s.

Inference. We use a residual bootstrap with $B = 1{,}000$ draws that keeps the observed weight path fixed and re-estimates both blocks in each draw. Differences $\Delta_h = \Phi_h(2019) - \Phi_h(2000)$ are formed draw by draw, because two overlapping bands do not test a difference. Bands are Hall (1992) intervals, $[\, 2\hat{\theta} - q^{*}_{1-\alpha/2},\ 2\hat{\theta} - q^{*}_{\alpha/2} \,]$, at 68% coverage. Hall intervals correct for the first-order bias of persistent levels VARs (Kilian, 1998). For H3 we compute likelihood-ratio statistics $LR = T\,(\ln|\hat{\Sigma}_R| - \ln|\hat{\Sigma}_U|)$ and obtain bootstrap p-values by regenerating data from the restricted model along the observed weight path. Chi-square p-values over-reject with persistent regressors.

4.3 Kernel time-varying SVAR (M2)

As an agnostic check, every coefficient is allowed to drift smoothly. Following Giraitis, Kapetanios and Yates (2014), at each date τ each block is estimated by weighted least squares,

$$\hat{\beta}_\tau = \Big(\sum_t k_{t\tau}\, X_t X_t'\Big)^{-1} \sum_t k_{t\tau}\, X_t\, y_t, \qquad k_{t\tau} = K\!\left(\frac{t-\tau}{H}\right), \tag{6}$$

with a Gaussian kernel K and bandwidth $H = T^{0.7}$ (47 months), using $p = 2$ and a local constant. Each block keeps its own regressors, so block exogeneity holds at every τ. Bandwidth $T^{0.5}$ leaves fewer effective observations at the sample ends than regressors, so the estimator stops unless the Kish effective sample exceeds twice the number of regressors. If the trade-channel story is right, the local impact loadings of Brazilian output on each partner should co-move with that partner's trade weight.

4.4 TVP-VAR with stochastic volatility (M3)

The final rung is the Primiceri (2005) model:

$$\begin{aligned} y_t &= X_t'\beta_t + A_t^{-1}\Sigma_t\,\varepsilon_t, & \beta_t &= \beta_{t-1} + \nu_t, \\ a_t &= a_{t-1} + \zeta_t, & \log\sigma_t &= \log\sigma_{t-1} + \eta_t, \end{aligned} \tag{7}$$

estimated by Gibbs sampling. The steps are a precision-based sampler for $\beta_{1:T}$ and $a_{1:T}$ (Chan and Jeliazkov, 2009), the seven-component mixture of Kim, Shephard and Chib (1998) for the log-volatilities, and the corrected draw order of Del Negro and Primiceri (2015). Block exogeneity holds in every draw: the state vector has no element for domestic lags in foreign equations, and the lower-triangular A_t with foreign variables first already zeros the foreign-on-domestic block. Because T is short, M3 uses a four-variable system: U.S. and Chinese activity, and Brazilian IP and CPI as 12-month changes. It uses $p = 2$ and 10,000 retained draws. Priors follow Primiceri ($k_Q = 0.01$, $k_S = 0.1$, $k_W = 0.01$), centred on full-sample OLS because no pre-1999 training sample comes from the same monetary regime.

4.5 Pre-tests: integration, cointegration and exogeneity

Before estimation, a screen tests each variable's order of integration (ADF and KPSS), the cointegration rank (Johansen, 1991), and block exogeneity (H4). Standard Wald tests of zero lag restrictions over-reject in levels VARs with integrated variables. Following Toda and Yamamoto (1995), we therefore estimate a VAR of order $p + d_{\max}$ and test only the first p domestic lags in the foreign equations, with bootstrap critical values (Hacker and Hatemi-J, 2006) and Holm adjustment across candidate splits. Weak exogeneity is tested on error-correction loadings (Pesaran, Schuermann and Weiner, 2004). Contemporaneous ordering cannot be tested with second moments alone; DirectLiNGAM (Shimizu et al., 2011) on non-Gaussian residuals is reported as advisory evidence.

5 Results

5.1 Pre-tests

Every variable except U.S. activity is I(1) by ADF and KPSS. The Johansen trace test finds a cointegrating rank of 1 out of 8. Information criteria choose $p = 6$ (AIC), 5 (HQ) and 4 (BIC). We use $p = 4$ as the headline and report $p = 2$ and $p = 6$ as robustness checks.

Table 3. Block exogeneity of the foreign block (H4): bootstrap lag-augmented Wald test that Brazilian variables do not Granger-cause the foreign block, 1999:07–2019:12.

Lags	Selected by	Bootstrap p	Holm p	Verdict
2	—	0.029	0.040	rejected
4	BIC	0.370	0.792	not rejected
6	AIC	0.263	0.533	not rejected

At $p = 4$ and $p = 6$ the verdict is "mixed": the federal funds rate fails weak exogeneity (it loads on the cointegrating relation).

Block exogeneity is rejected at $p = 2$ but not at $p = 4$ or $p = 6$ (Table 3). The rejection at two lags reflects residual autocorrelation leaking into the Granger test rather than genuine feedback from Brazil to the world economy. DirectLiNGAM places Chinese activity first and agrees with the foreign-first ordering for 81% of foreign–domestic pairs. One weakness remains: the multivariate portmanteau test rejects white-noise residuals at $p = 4$ ($p < 0.001$, $h = 12$), probably because of seasonality in the unadjusted CPI and the 2008–09 crisis.

5.2 Is the time variation real?

Table 4. Tests of parameter constancy (M0) and of trade-driven transmission (M1), $p = 4$. Bootstrap p-values re-estimate both models on data regenerated under the null.

Null hypothesis	Statistic	df	Bootstrap p
M0: no break at 2009:01 (Chow), domestic block	491.7	196	0.010
M0: no break at any date (sup-LR; strongest at 2004:03)	758.8	196	0.005
M1: weights do not change transmission ($G = 0$)	91.8	40	0.005
M1: GVAR star loading is sufficient	104.5	60	0.070

The constant-coefficient model is rejected both at the 2009 date and at an unknown break date (Table 4). Letting transmission depend on trade weights improves the fit significantly ($p = 0.005$), which supports H3. The star restriction is borderline ($p = 0.07$): a single shared slope comes close to explaining the change, but partner-specific slopes are preferred. A Wald test of equal U.S. and China slopes rejects (statistic 52.1, 20 df, $p < 0.001$).

5.3 Headline impulse responses

Table 5. Response of Brazilian industrial production (%) to one-standard-deviation foreign activity shocks, with trade weights frozen at each benchmark year. M1, interacted, $p = 4$, 1,000 bootstrap draws, 68% Hall bands for the difference.

Shock	Month	2000 weights	2009 weights	2019 weights	2019 − 2000 [68% band]
China activity	0	+0.05	+0.16	+0.36	+0.31 [0.09, 0.60]
China activity	6	−0.30	+0.38	+1.31	+1.61 [1.54, 2.20]
China activity	12	+0.45	+0.75	+1.37	+0.92 [0.87, 1.76]
China activity	24	+0.24	+0.16	+0.88	+0.64 [0.30, 1.34]
U.S. activity	0	−0.17	+0.13	+0.22	+0.39 [0.27, 0.56]
U.S. activity	6	+0.42	+0.46	+0.64	+0.22 [0.01, 0.41]
U.S. activity	12	+0.37	+0.09	+0.16	−0.21 [−0.50, 0.06]
U.S. activity	24	+0.03	−0.32	−0.25	−0.27 [−0.61, 0.01]

China (H1). With 2019 trade weights, a Chinese activity shock raises Brazilian industrial production by 1.80% at its peak in month 9. With 2000 weights the peak is 0.69%, in month 16. The largest difference is +1.93 percentage points at month 8, with a 68% band of [1.92, 2.70], and the difference band excludes zero from month 6 through month 24 (Table 5, Figures 1–2). H1 is supported.

United States (H2). The U.S. share fell from 22% to 14%, yet the peak response of Brazilian output to U.S. activity is about the same at 2000 weights (0.53%) and 2019 weights (0.77%); it only fades faster after a year. This rejects the trade-only version of H2. It matches Cesa-Bianchi et al. (2012), who find that the U.S. effect on Latin America works through more than bilateral trade, for example through financial conditions and third-country linkages.

Prices and the exchange rate. At 2000 weights, Brazilian CPI rises after a Chinese activity shock (+0.65 at month 12), consistent with a commodity-price channel. At 2019 weights it falls (−0.33). At the same horizon the real appreciates by 2.35% with 2019 weights. One interpretation is that a Chinese boom now strengthens the real enough to lower import prices. This is a reading of the point estimates, not a tested result.

A note on bands. In a few cells the Hall band lies slightly to one side of the point estimate (2019 weights, month 12: +1.37 against [1.41, 2.15]). This signals large small-sample bias in a levels VAR of this size. A bias-corrected bootstrap-after-bootstrap (Kilian, 1998) is the natural next step.

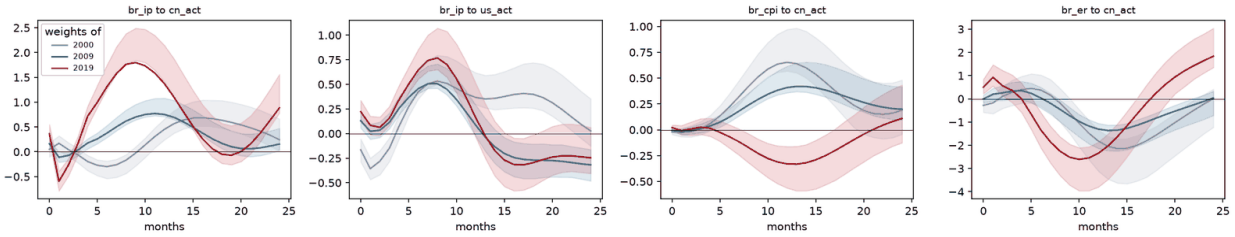


Figure 1. Frozen-weight impulse responses (M1, interacted, $p = 4$) of Brazilian industrial production to Chinese and U.S. activity shocks, and of Brazilian CPI and the exchange rate to Chinese activity shocks. Grey: 2000 weights; blue: 2009; red: 2019. Shaded areas are 68% Hall bands from 1,000 bootstrap draws.

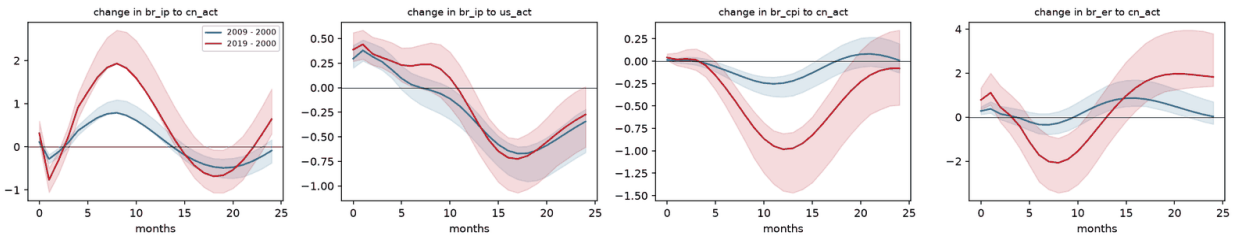


Figure 2. Differences between frozen-weight responses (2009 − 2000 and 2019 − 2000), computed draw by draw, with 68% Hall bands.

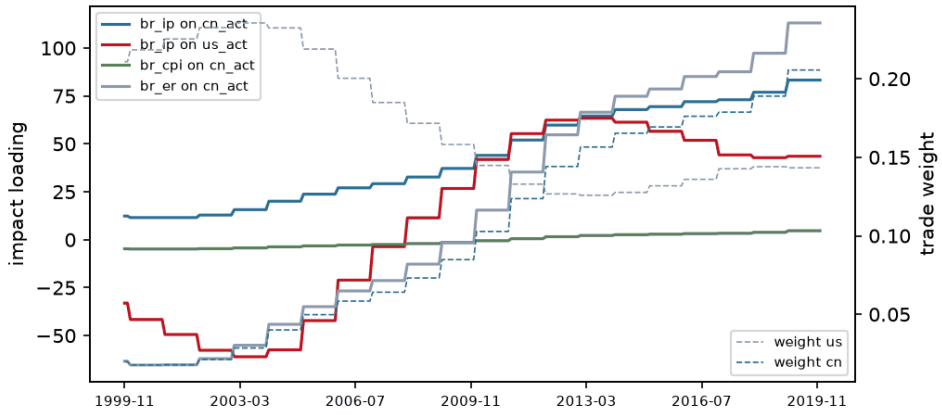


Figure 3. Impact loadings of Brazilian variables on partner activity implied by the estimated M1 coefficients and the observed weight path, 1999:11–2019:12 (left axis), with the U.S. and China trade weights (right axis, dashed).

6 Robustness

6.1 Lag order and the Cesa-Bianchi et al. benchmark

Table 6. Peak response of Brazilian industrial production to a Chinese activity shock, and the bootstrap LR test of trade-driven transmission, by lag order.

Lags	Peak, 2000 weights	Peak, 2019 weights	LR bootstrap p
2	0.55	1.45	0.015
4	0.69	1.80	0.005
6	0.44	1.89	0.020

The China result holds at all three lag orders (Table 6). Restricting the sample to end in 2009:12, as Cesa-Bianchi et al. (2012) did, and comparing 1995 with 2009 weights, the mean response over months 0–24 rises by +0.56 for China and falls by −0.10 for the United States. The direction matches the published GVAR result. At $p = 2$ the China change is positive over months 0–12 and slightly negative over months 12–24, so the long-horizon part depends on lag length.

6.2 Kernel estimator (M2)

Brazil's local impact loading on Chinese activity rises with China's trade weight (correlation +0.44). The loading on U.S. activity moves against the U.S. weight (correlation −0.88): it becomes less negative as the

U.S. share falls (Figure 4). The unrestricted drift therefore supports the China story and indicates that the change in U.S. transmission is not a trade-share effect, consistent with the M1 evidence on H2.

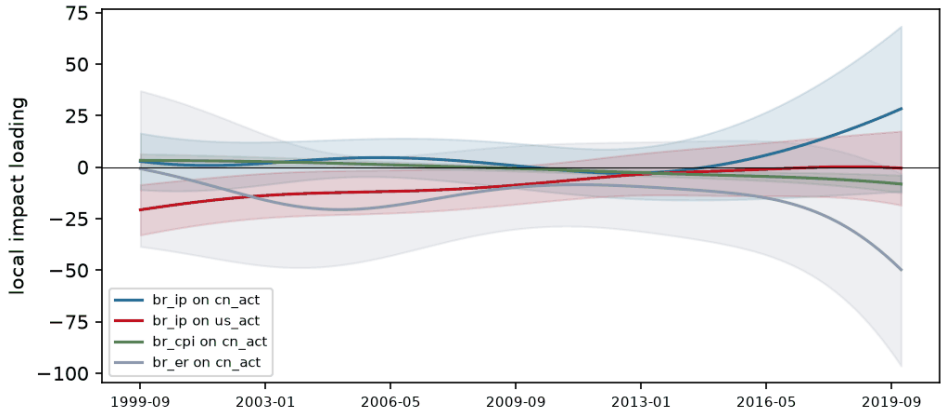


Figure 4. Kernel-estimated (M2) local impact loadings of Brazilian industrial production on Chinese and U.S. activity, with each partner's trade weight. Gaussian kernel, bandwidth $T^{0.7} = 47$ months, $p = 2$.

6.3 TVP-VAR with stochastic volatility (M3)

Responses barely move between 2002, 2009 and 2019 in M3. The peak response of IP growth to Chinese activity at month 12 is 0.44, 0.46 and 0.32 respectively, with overlapping posterior bands. The change appears in volatility instead: Brazilian IP shocks are largest in late 2009 and in 2018–19 (Figure 5). Primiceri's priors allow very little coefficient drift ($k_Q = 0.01$), and without a training sample they are centred on full-sample OLS, so M3 is conservative by construction. Convergence is adequate but not clean: the maximum Geweke $|z|$ for the coefficient block is 4.3. We read M3 as a lower bound on coefficient drift rather than as evidence against H1.

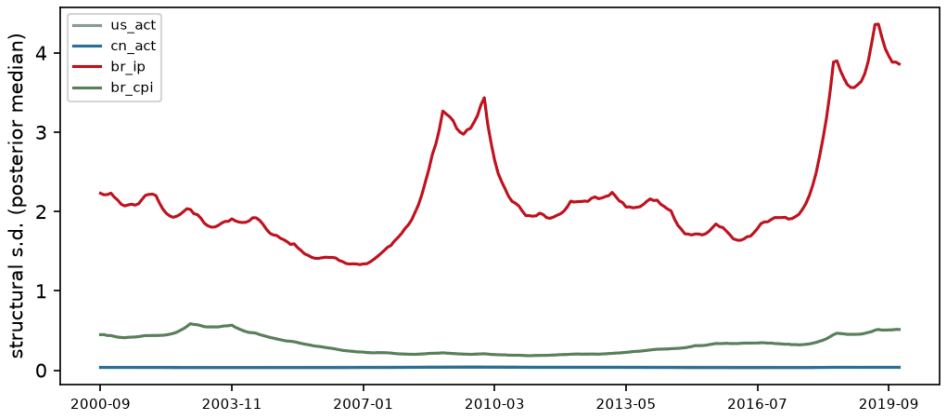


Figure 5. Posterior median structural shock standard deviations from the TVP-VAR with stochastic volatility (M3).

6.4 Sample through 2023

Adding 2020–2023 breaks M1. Impact loadings jump to implausible values (an impact response of +16.6 to a China shock at 2000 weights), because the 2020 collapse dominates a regression with interaction

terms. The constancy test still rejects (bootstrap $p = 0.005$), but the responses are not usable without pandemic dummies, so the headline sample ends in 2019.

7 Validation of the estimators

Before any Brazilian result was produced, each estimator had to pass cases of three kinds, defined as data files with SHA-256-pinned fixtures and stated pass rules (`replication/cases/`). The full suite passes 18 of 18 cases; the fast suite of 13 runs in continuous integration.

Table 7. Selected replication cases.

Kind	Check	Result
Parity	M0 and M1 ($G = 0$) coefficients against `statsmodels` VAR	agree to 10^{-8}
Parity	M3 conditional posterior mean of coefficient states against the Kalman smoother	agree to 10^{-14}
Parity	M2 with a flat kernel against the constant estimator	agree to 10^{-8}
Simulation	M1 90% band coverage of true frozen-year IRFs (200 reps, $T = 400$)	0.895 / 0.880 / 0.857 (first year / last year / difference)
Simulation	M1 LR test size and power at 5%	size 0.046, power 1.0
Simulation	M3 90% band coverage of coefficient / log-variance paths	0.83 / 0.68 (volatility bands under-cover)
Published	Primiceri (2005): U.S. interest-rate volatility peak	posterior median peaks in 1980Q4
Published	Cesa-Bianchi et al. (2012): direction of China and U.S. changes	reproduced (qualitative)

8 Limitations

- **Endogenous weights.** Trade shares respond to the same shocks the model studies. Lagging and smoothing the weights reduces this problem but does not remove it.
- **Activity measure.** The OECD cycles are detrended indices. Responses are to one-standard-deviation moves in those cycles, not to GDP growth.
- **Weak exogeneity.** The federal funds rate fails weak exogeneity, so the foreign block is "mixed" on statistical grounds.
- **Residual autocorrelation.** Some autocorrelation remains at $p = 4$ (portmanteau $p < 0.001$).
- **Bands.** The 68% bands are narrower than conventional 90% or 95% intervals, and Hall intervals show signs of bias. Kilian's bias-corrected bootstrap is the next step.
- **M3 priors.** The M3 prior is data-dependent, and its volatility bands under-cover in simulation.

9 Reproducibility

All code is in the open-source Python package `csuf-fullerton` at https://github.com/espin086/csuf-fullerton. The results in this paper come from commit `4b832fc` on branch `feat/0005-time-varying-svar` (pull request #18). Key sources:

- Trade weights: `src/csuf_fullerton/core/tvsvar/weights.py`

- M0–M3 estimators: `src/csuf_fullerton/core/tvsvar/` (`m0.py`, `m1.py`, `m2.py`, `m3.py`, `precision.py`)
- Block-exogenous SVAR, bootstrap and exogeneity screen: `src/csuf_fullerton/core/svar/`
- Model design and implementation decisions: `openspec/changes/0005-time-varying-svar/design.md`
- Data and theory specifications: `examples/brazil/`
- Full outputs (tables, figures, settings and data hashes in `run.json`): `reports/brazil/`
- Reproduction script: `scripts/reproduce_brazil.sh`

```
make install && source .venv/bin/activate
csuf-fullerton replicate --suite fast       # estimator checks (about one minute)
make brazil-check                           # rebuild every Brazil output and compare
```

10 Conclusion

When a small open economy changes trading partners, the transmission of foreign shocks changes with it, and a constant SVAR misses this. A trade-weighted block-exogenous SVAR shows that Brazil's industrial production has become roughly two and a half times more responsive at its peak to Chinese activity shocks as China's trade share rose from 2% to 21%. Formal tests reject constant transmission, and the foreign block passes exogeneity tests at adequate lag lengths. The U.S. effect has not declined with the U.S. trade share, which points to financial channels. A kernel estimator agrees on China; a Bayesian TVP-VAR with tight priors places the change in volatility. Natural extensions are bias-corrected bands, a shadow federal funds rate for the zero-lower-bound period, pandemic dummies to extend the sample, and other commodity exporters whose trade has shifted toward China, such as Chile and Peru.

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